

VARUN SINGHAL

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EDUCATION

University of Southern California	Los Angeles, CA
Master of Science in Computer Science: Artificial Intelligence	January 2025 – December 2026
Concentration Focus: Artificial Intelligence, Machine Learning, Quantum Computing + Cryptography	
University of California, San Diego	San Diego, CA
Bachelor of Science: Computer Science	October 2020 – June 2024
GPA: 3.83/4.00, Provost Honors (2020 – 2024)	

PROFESSIONAL EXPERIENCE

GEICO	San Francisco, CA
Software Engineer	May 2024 – August 2025
- Optimized a production quote orchestration and estimation system, reducing end-to-end latency from ~30s to <13s and compute costs by 50%+ while meeting a <15s SLA and preserving quote consistency. - Re-architected the critical path by eliminating unnecessary service calls, parallelizing independent requests, reducing duplicate computation, and deferring non-critical processing that did not affect initial pricing decisions. - Led a cross-team ML scoring-model initiative from feature design through production integration to improve premium-estimation accuracy across partner quoting workflows. - Productionized ML and backend services using AWS SageMaker, Lambda, DynamoDB, REST APIs, and CI/CD, improving reliability and supporting high-volume production inference. - Authored API documentation, technical guides, and video tutorials to speed onboarding across 5+ engineering teams.	
Amazon Web Services (AWS)	Seattle, WA
Software Engineering Intern	May 2023 – August 2023
- Worked directly with Fortune 500 customers to design and provision production database and storage pipelines, integrating new workloads with existing enterprise environments and onboarding 5 customer workloads. - Built automated CloudWatch and GuardDuty health checks that reduced failure-detection time by ~40%.	

RESEARCH & TECHNICAL PROJECTS

VLM4Rec: Multimodal Semantic Representation for Recommendation with Large Vision-Language Models	arXiv, March 2026
Python, PyTorch, LLaVA-NeXT 7B, Sentence-BERT, CLIP, Hugging Face, NVIDIA GPU	
- Built an end-to-end multimodal recommender system using LLaVA-NeXT 7B semantic descriptions, Sentence-BERT item embeddings, and cosine-similarity retrieval over user histories. - Designed and optimized a GPU inference pipeline using 4-bit quantized LLaVA-NeXT 7B on an NVIDIA RTX 5080, batching 4,708 catalog items and precomputing/caching semantic embeddings to shift expensive inference offline. - Evaluated 12 text, vision, and multimodal fusion approaches across 2,914 users; achieved 0.3539 Recall@10 and 0.2745 NDCG@10, improving Recall@10 by 54.9% over the BERT baseline.	
Low Latency C++ Arbitrage Engine C++	
Implemented a graph-based C++ arbitrage engine and real-time market-data pipeline across Binance, Coinbase, and Kraken with <10ms detection and <300ms execution latency.	
Prediction Market Arbitrage Scanner Python, FastAPI	
Built a real-time pipeline to ingest, normalize, semantically match, and validate Polymarket/Kalshi markets; event-level filtering reduced LLM calls by ~80% while minimizing false-positive matches.	

RELEVANT SKILLS

- Languages:** C/C++, Python, SQL
- GPU & AI:** CUDA C/C++, CUDA Kernels, Parallel Programming, GPU Programming, NVIDIA GPUs, NVIDIA Nsight, Machine Learning, Deep Learning, AI Algorithms, PyTorch, TensorFlow, Hugging Face Transformers, LLaVA-NeXT, Sentence-BERT, CLIP, Vision-Language Models, Recommender Systems, Computer Vision, NLP
- Systems & Performance:** Linux, Computer Architecture, Software Design, Algorithms, Low-Level Performance Optimization, Model Inference, Quantization, Information Retrieval, Latency & Throughput Optimization, Performance Profiling, AWS SageMaker, CI/CD